Ethanol Blending in Gasoline – Malaysia

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U.S. GRAINS & BIOPRODUCTS COUNCIL

20 F St. NW, Suite 900 \ Washington, DC 20001
202-789-0789 \ www.grains.org

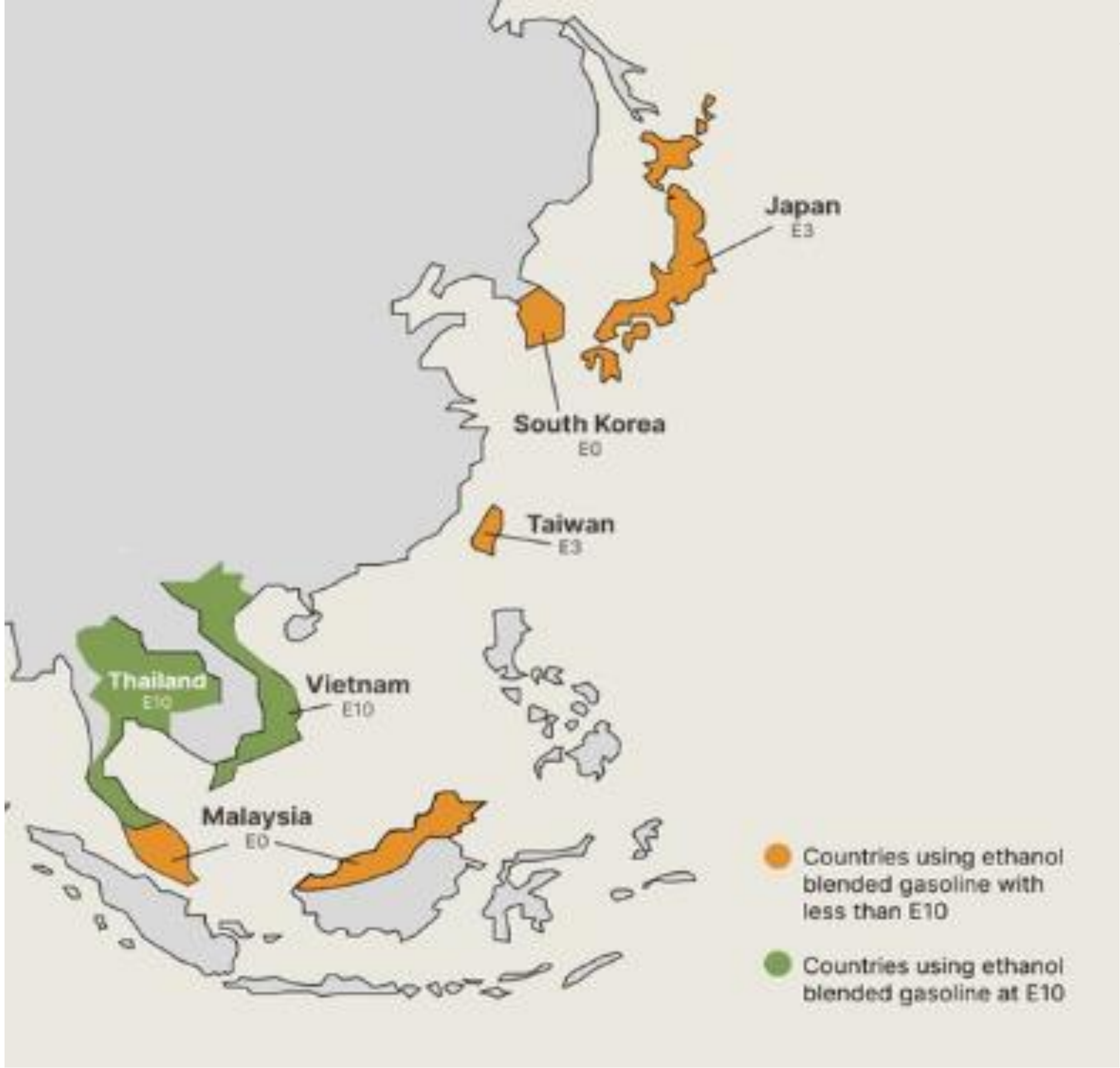
FARO90 HCX

www.faro90.com
+52 55 6122 2255

Ethanol Blending in Asia

There are important fuel quality and environmental impact of vehicle emission challenges in these regions.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Asia to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Seven countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

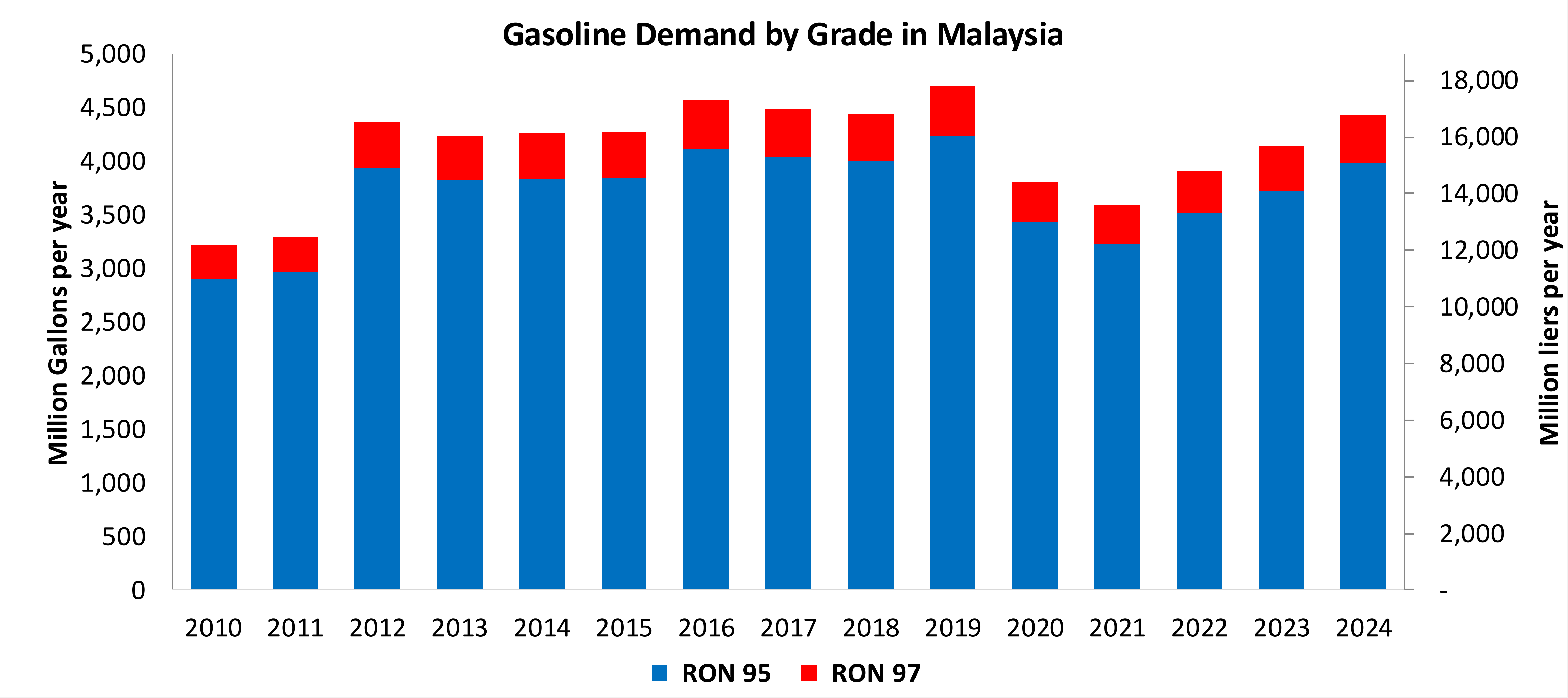
Ethanol Gasoline Blending in Malaysia

In 2024, gasoline consumption in Malaysia was 16,784 million liters (4,434 million gallons) and growth rate was -1.2%. Gasoline production and net imports were 5,436 and 11,348 million liters (1,436 and 2,998 million gallons) correspondingly. Malaysia offers two main grades: RON 95 and RON 97) with an average octane of 95.2 RON.

Government support for biofuels has so far targeted biodiesel. The Biofuel Industry Act of 2007 excluded ethanol as the source of alternative fuels under the National Biofuels Policy of 2006. As there is no policy support for ethanol, there is no production of fuel ethanol and minimal consumption. In 2024, ethanol consumption in Malaysia was 16 million liters (4 million gallons) representing 0.1% of gasoline demand.

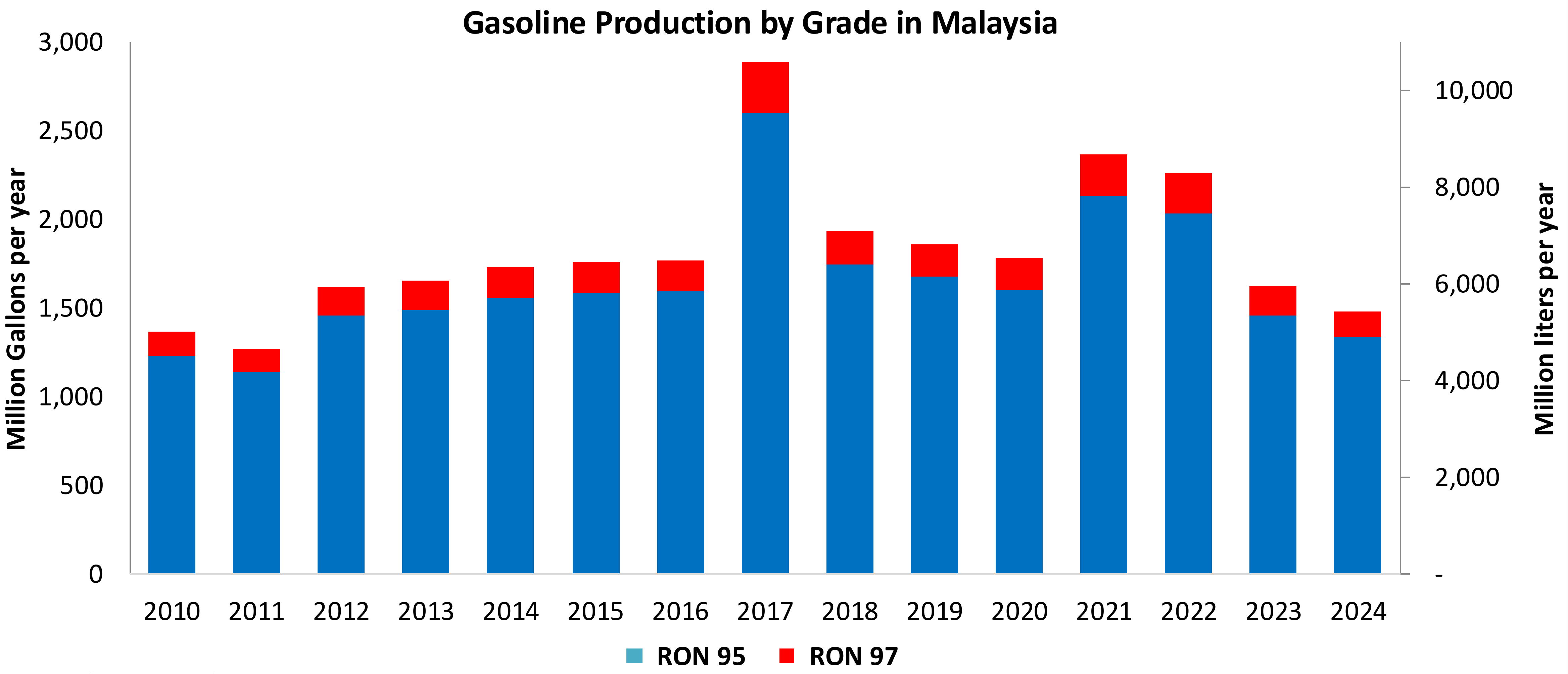
Source: EIA, Suruhanjaya Tenanga, USDA, USGrains

Gasoline Demand in Malaysia



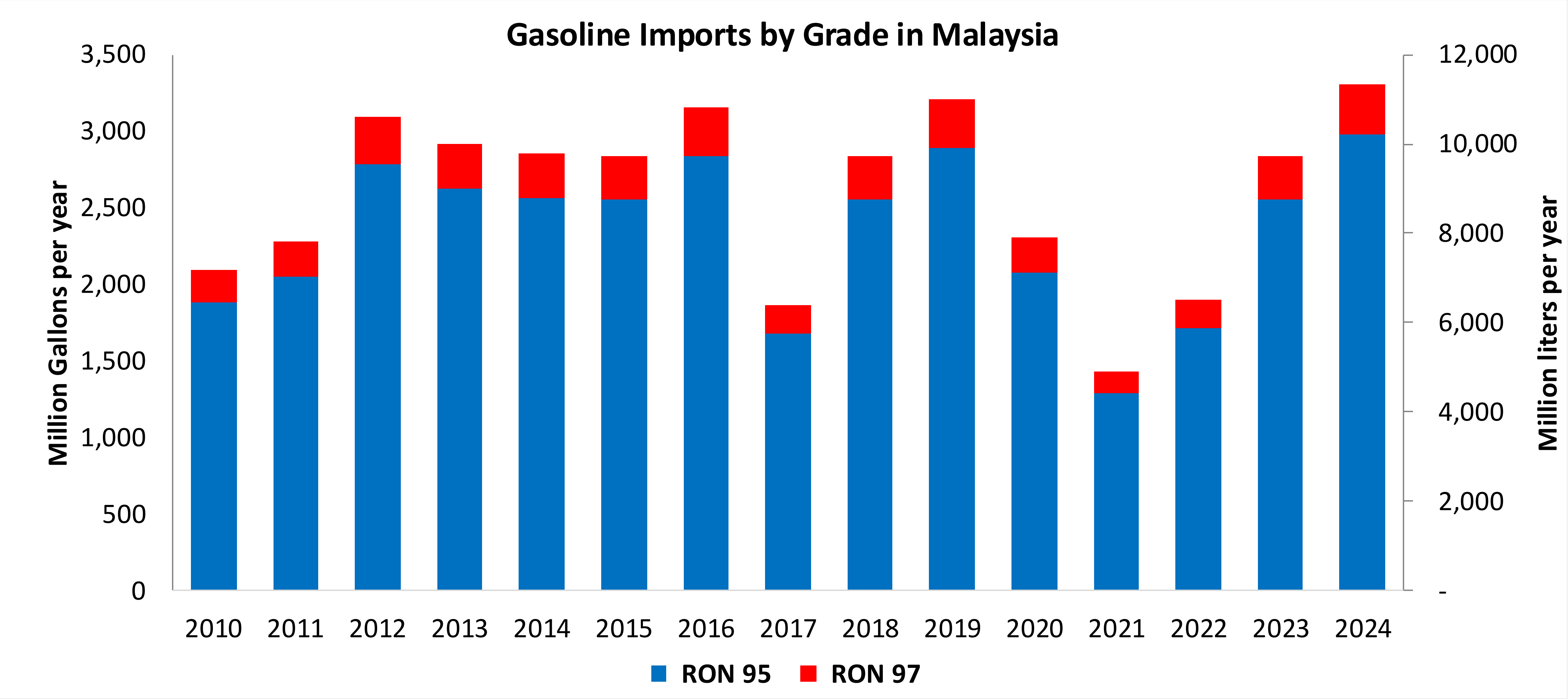
Source: EIA, Suruhanjaya Tenanga

Gasoline Production in Malaysia



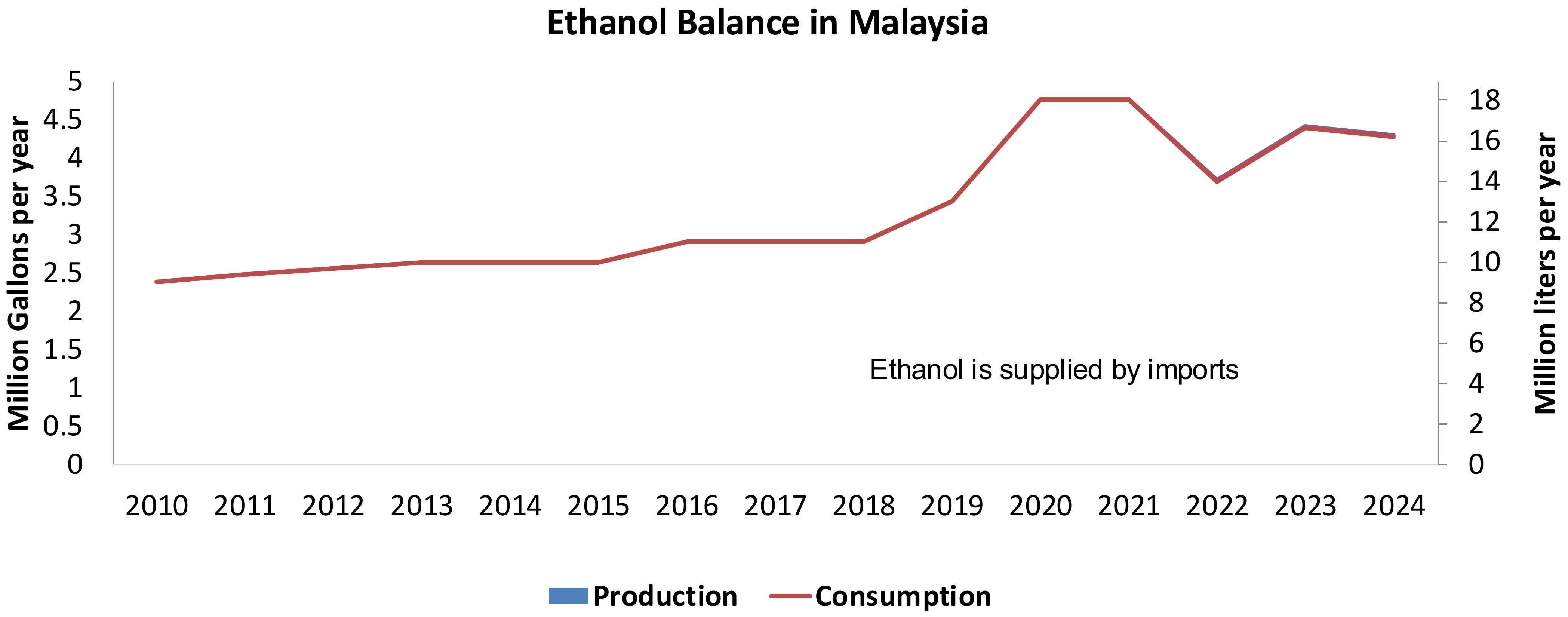
Source: EIA, Suruhanjaya Tenanga

Gasoline Net Imports in Malaysia



Source: EIA, Suruhanjaya Tenanga

Ethanol Balance in Malaysia



Source: USDA, USGrains

Gasoline Quality in Malaysia

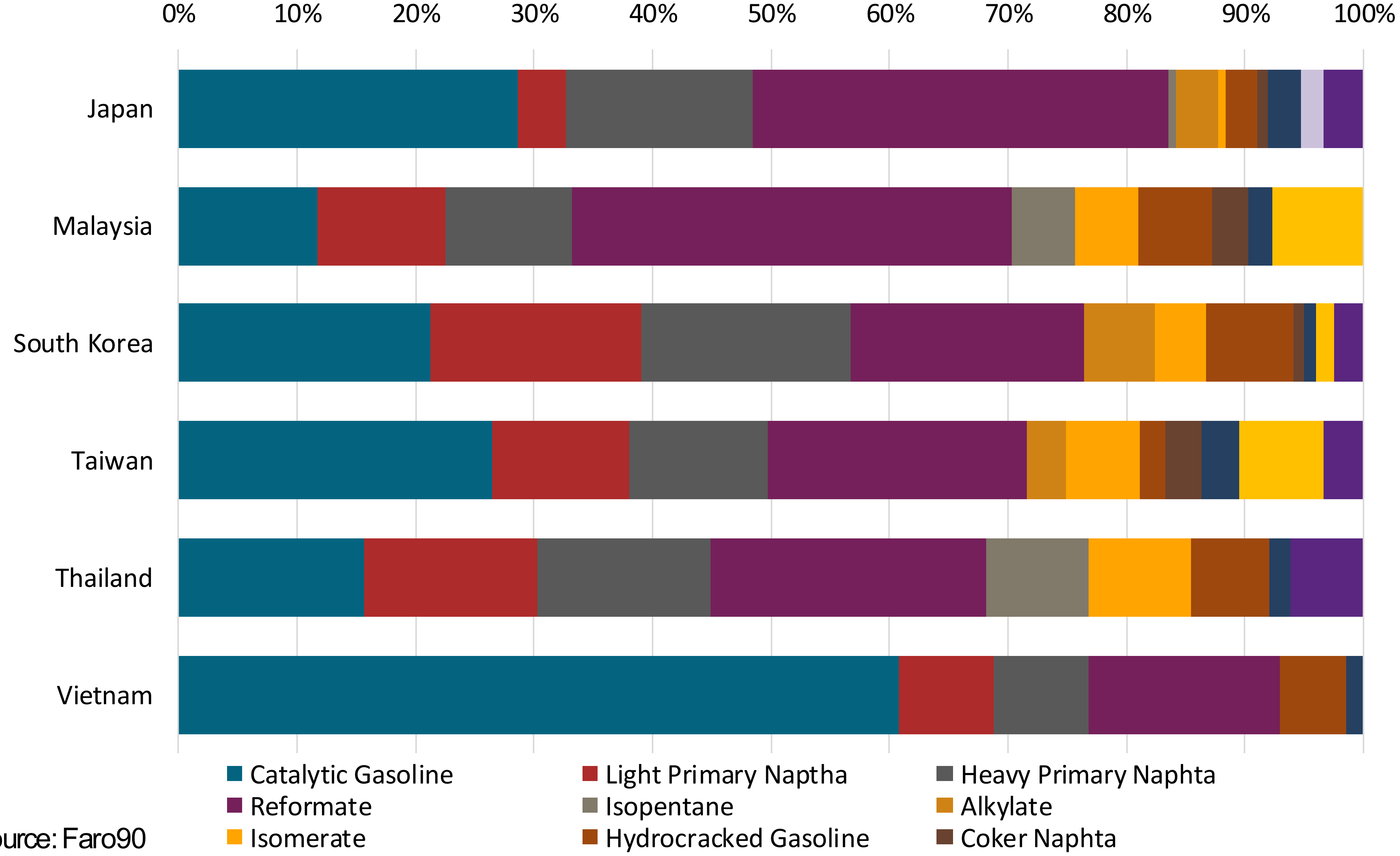
Name	MS 118-3:2011			
Implementation Year	2011			
Applicability	Nationwide			
Grade	RON 95		RON 97	
	Min	Max	Min	Max
RON	95.0		97.0	
MON				
AKI				
Benzene (%vol)		1.0		1.0
Aromatics (% vol)		35.0		35.0
Olefins (%vol)		18.0		18.0
Lead (g/L)		0.013		0.013
Manganese (mg/L)				
Sulfur (ppm)		50		50
Oxygen (%v)				
Ethanol (%vol)				
MTBE (%vol)				
RVP (psi)	60.0	70.0	60.0	70.0

Source: Department of Standards Malaysia, Euro 4 Spec

Gasoline Component Blending in Asia

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Asia are:

- Catalytic Gasoline
- Light Primary Naptha
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics



Source: Faro90

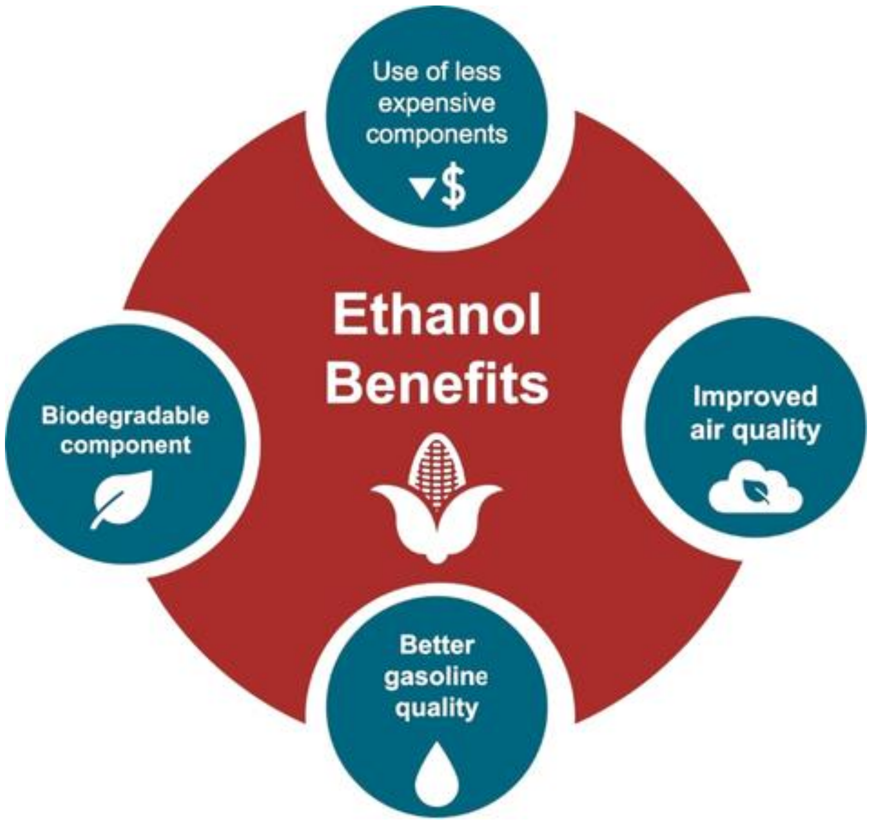
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

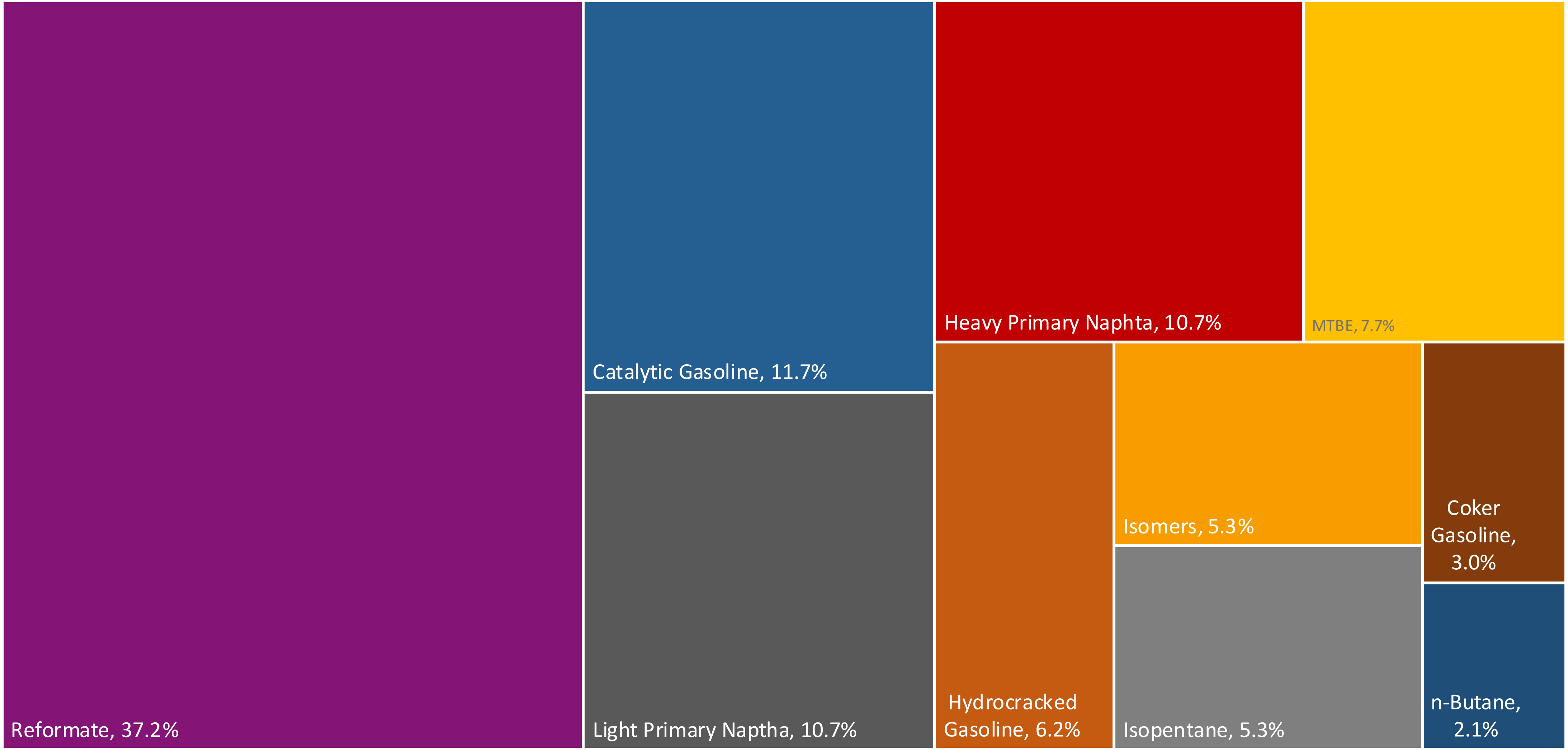
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

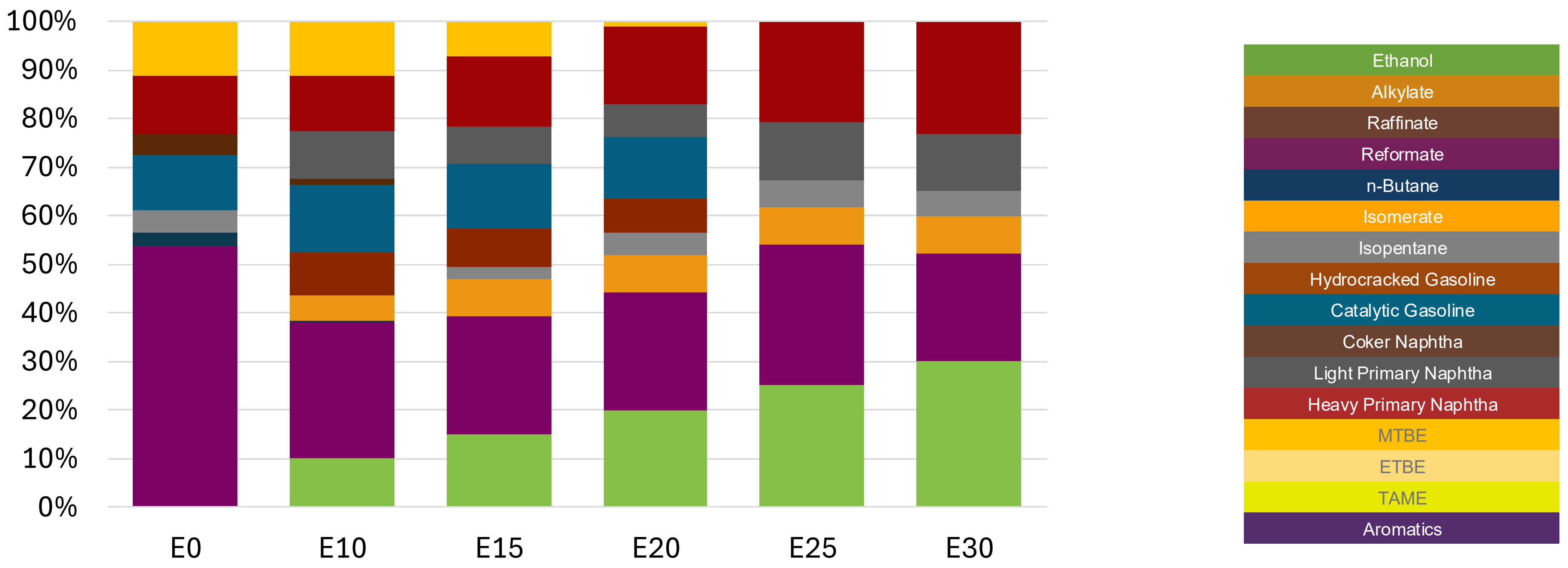


Malaysia Available Blending Components



Source: Faro90

Malaysia – Gasoline 95 – Constant Octane

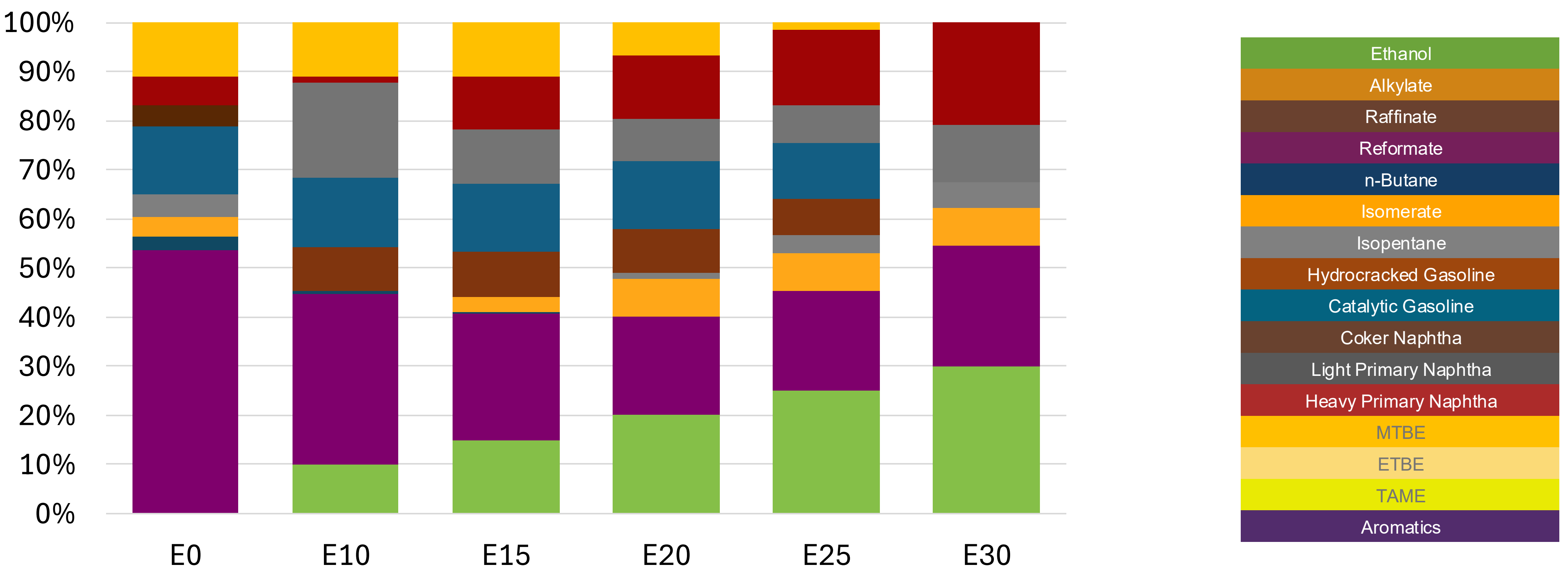


Octane (RON)	95.0	95.0	95.0	95.0	95.6	96.6
Price (US$/gal)	$ 2.360	$ 2.232	$ 2.164	$ 2.102	$ 2.025	$ 1.975

Average CIF 2024 Prices

Source: Faro90

Malaysia - Gasoline 97 - Constant Octane

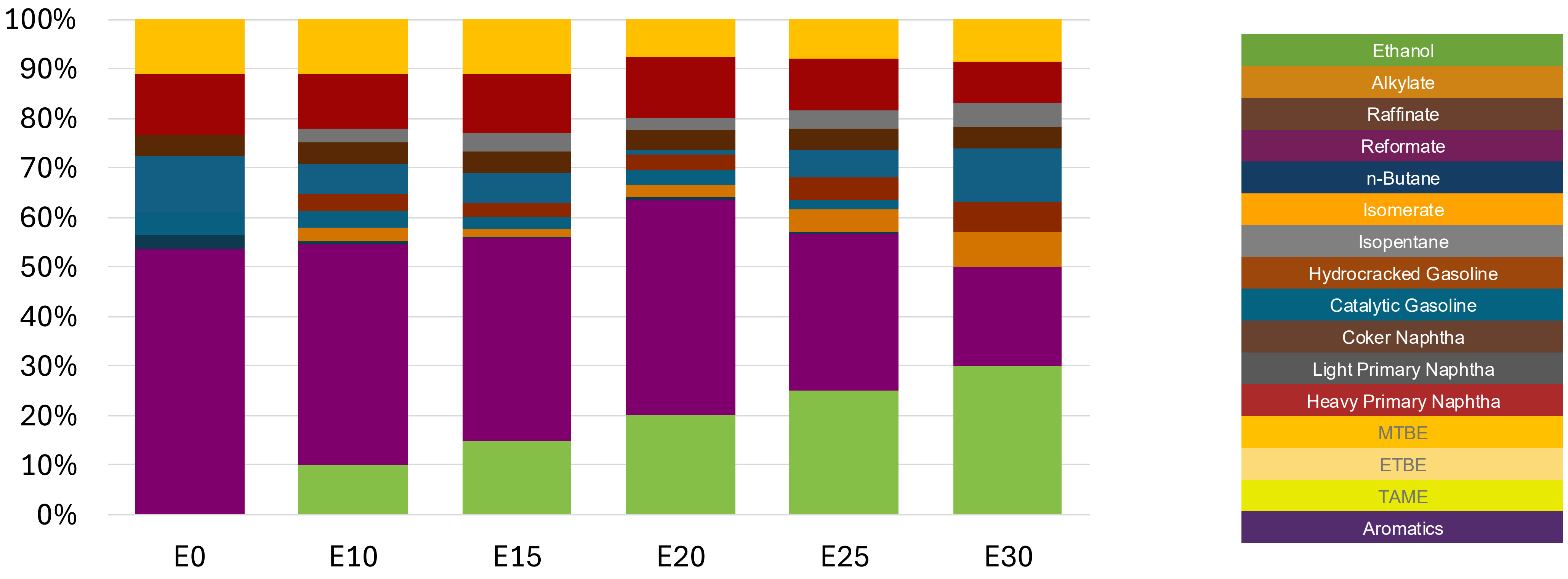


Octane (RON)	97.0	97.0	97.0	97.0	97.0	97.5
Price (US$/gal)	$ 2.466	$ 2.345	$ 2.249	$ 2.187	$ 2.116	$ 2.033

Average CIF 2024 Prices

Source: Faro90

Malaysia – Gasoline 95 – Increased Octane

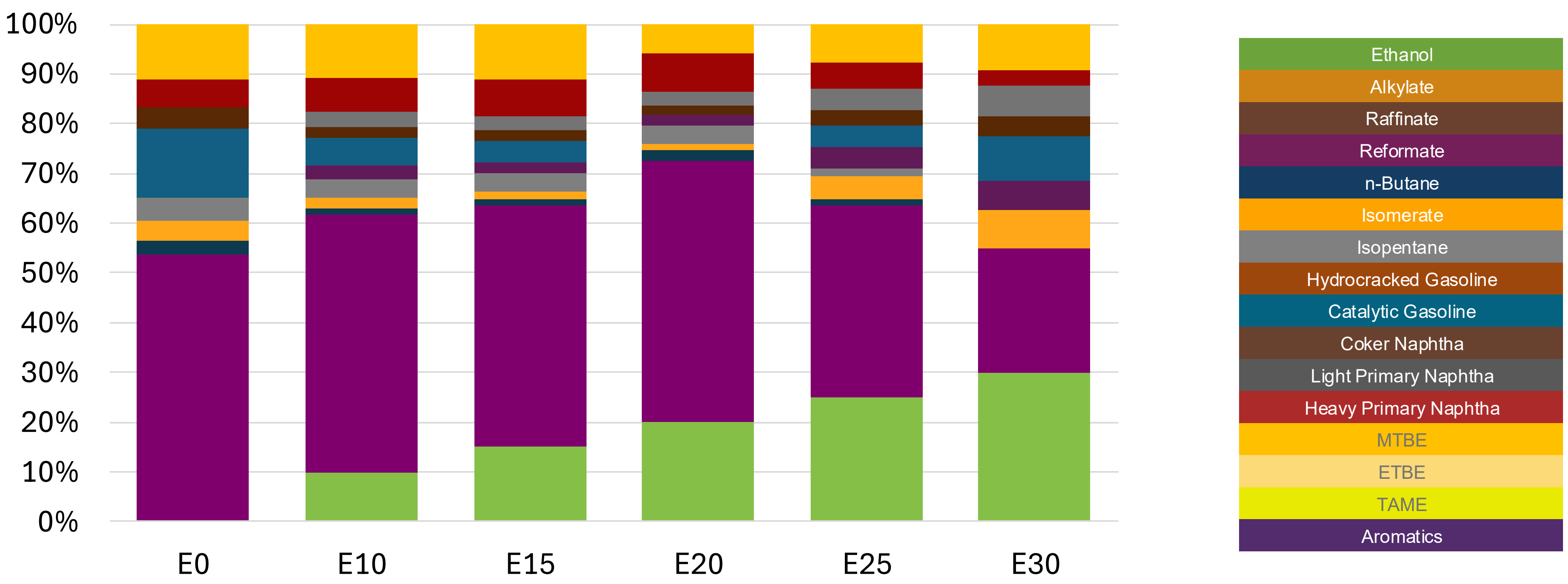


Octane (RON)	95.0	97.8	99.4	100.5	101.2	103.2
Price (US$/gal)	$ 2.360	$ 2.360	$ 2.360	$ 2.360	$ 2.360	$ 2.360

Average CIF 2024 Prices

Source: Faro90

Malaysia – Gasoline 97 – Increased Octane



Octane (RON)	97.0	99.8	101.3	102.1	102.9	104.9
Price (US$/gal)	$ 2.466	$ 2.466	$ 2.466	$ 2.466	$ 2.466	$ 2.466

Average CIF 2024 Prices

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model (IVE).]()
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

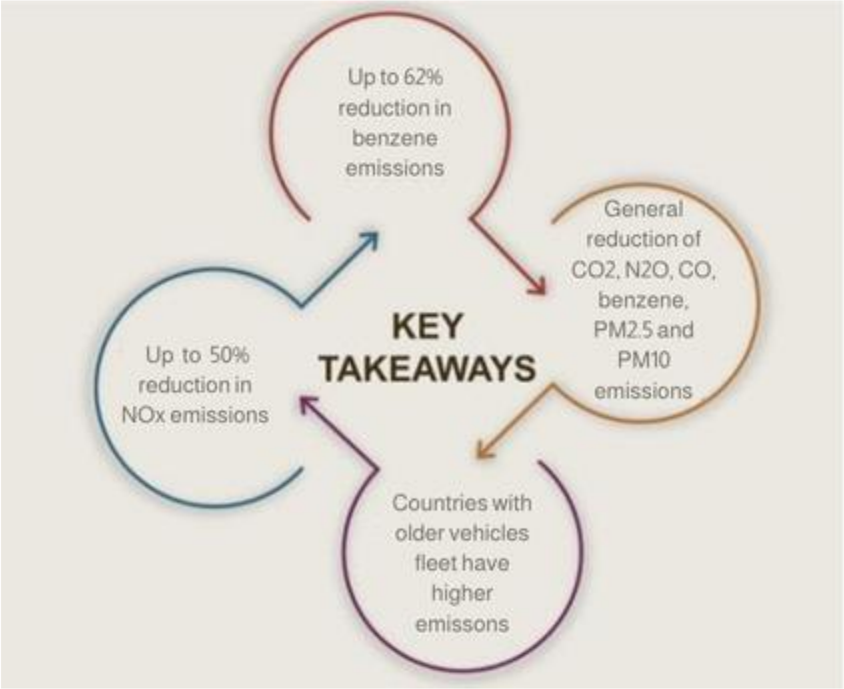
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

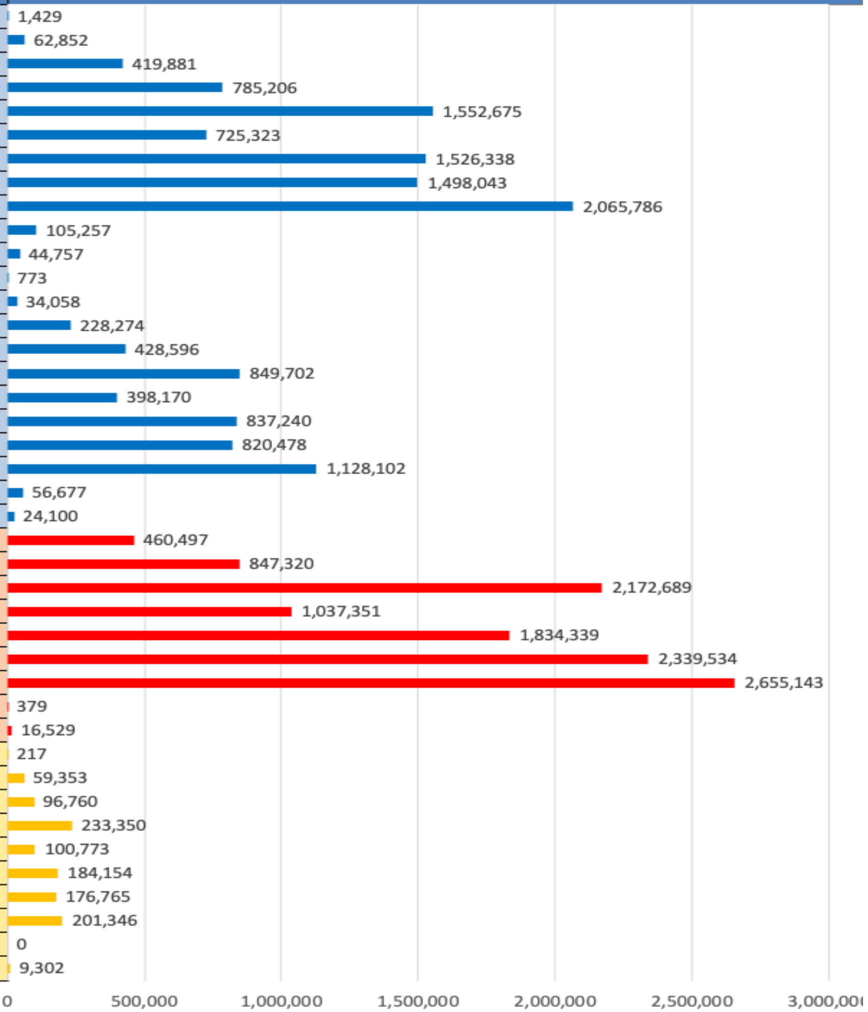
Source: Bureau of transportation statistics.

Main Results



Gasoline Vehicle Fleet - Malaysia

Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	1,429
		Single FI	3-Ways		>8 years	62,852
		Multi FI	3-Ways		>8 years	419,881
			Euro III		>8 years	785,206
			Euro IV		>8 years	1,552,675
			Euro V		>8 years	725,323
			Euro 6		>8 years	1,526,338
					4-8 years	1,498,043
					<4 years	2,065,786
			Hybrid		All	105,257
		Electric			All	44,757
	Medium	Carburator	3-Ways	PCV Tank	>8 years	773
		Single FI	3-Ways		>8 years	34,058
		Multi FI	3-Ways		>8 years	228,274
			Euro III		>8 years	428,596
			Euro IV		>8 years	849,702
			Euro V		>8 years	398,170
			Euro 6		>8 years	837,240
					4-8 years	820,478
					<4 years	1,128,102
			Hybrid		All	56,677
		Electric			All	24,100
Sml Engines	Light	4-Cycle	None	PCV	>8 years	460,497
			Euro III		>8 years	847,320
			Euro IV		>8 years	2,172,689
			Euro V		>8 years	1,037,351
			Euro 6		>8 years	1,834,339
					4-8 years	2,339,534
					<4 years	2,655,143
			Hybrid		All	379
		Electric			All	16,529
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	217
		Fuel-Injection	3-Ways		>8 years	59,353
			Euro III		>8 years	96,760
			Euro IV		>8 years	233,350
			Euro V		>8 years	100,773
			Euro 6		>8 years	184,154
					4-8 years	176,765
					<4 years	201,346
			Hybrid		All	0
		Electric			All	9,302

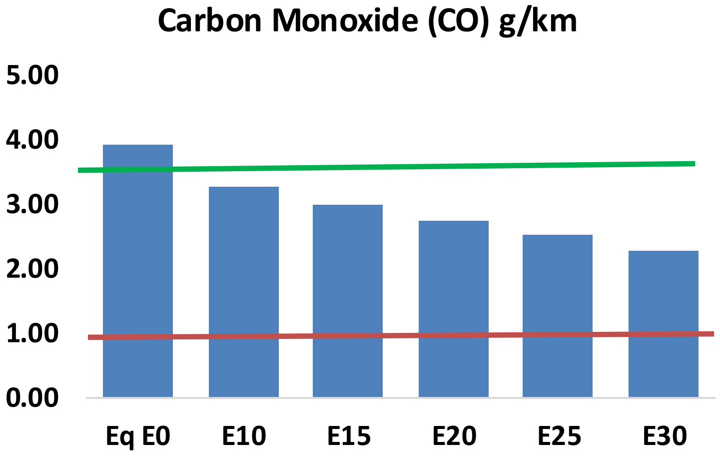


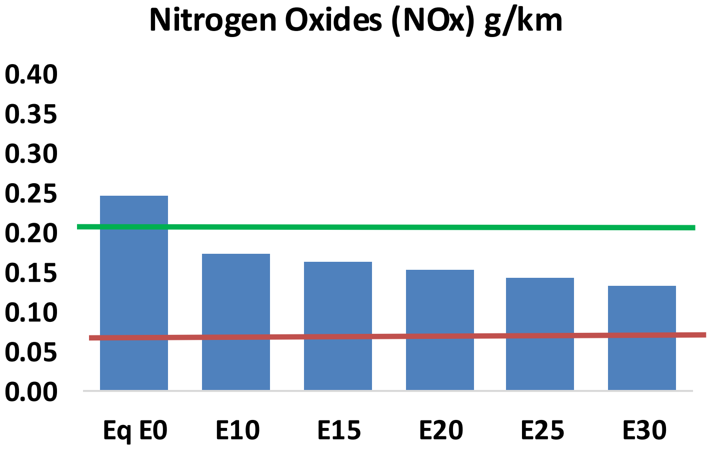
Vehicle Fleet: 26,019,519
Gasoline Vehicle Fleet: 22,285,447

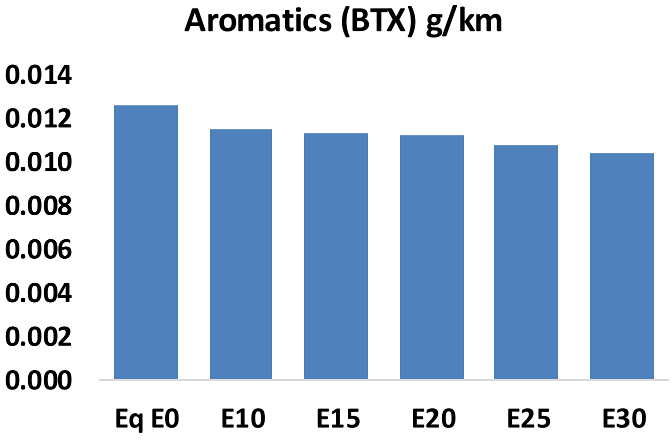
Source: Data Gov Malaysia

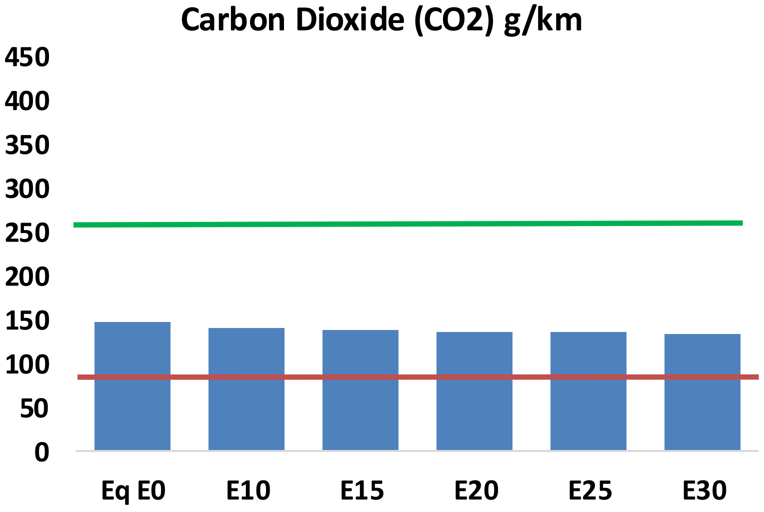
Malaysia – Vehicular Emissions

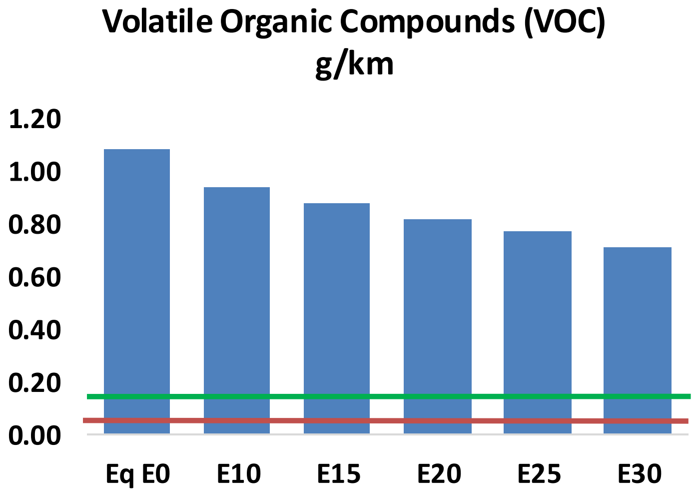
TIER III BIN 125 United States
Euro 6 Standard

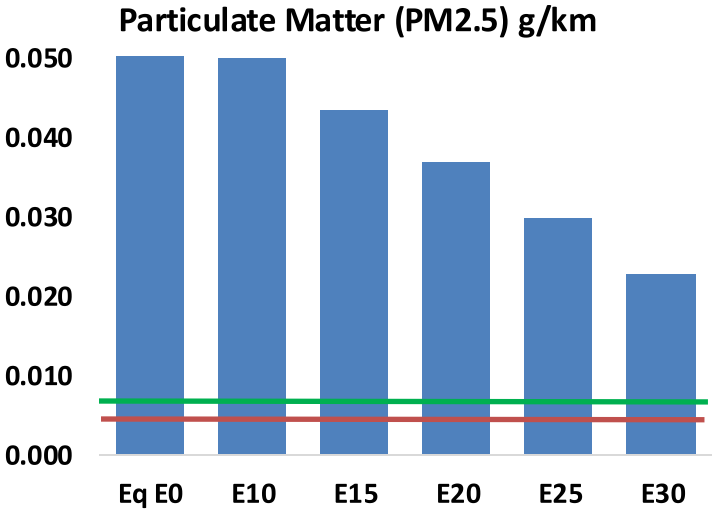












Source: Faro90

Malaysia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	2,144,339	1,967,241	1,790,144	1,642,629	1,493,082	1,377,756	1,247,742	-17%	-30%
CO2	80,754,602	78,678,055	76,716,872	75,174,459	74,411,212	74,065,375	72,700,137	-5%	-8%
VOC	594,330	553,078	511,826	478,887	446,139	421,077	387,209	-14%	-25%
NOx	135,754	101,816	95,028	89,564	84,466	78,846	72,900	-30%	-38%
SOx	1,829	1,691	1,553	1,436	1,324	1,202	1,074	-15%	-28%
NH3	36,345	35,986	35,626	35,795	36,197	36,481	36,716	-2%	0%
N2O	1,108	1,102	1,098	1,103	1,126	1,139	1,151	-1%	2%
CH4	130,499	130,499	130,499	132,457	135,328	137,807	140,156	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,805	2,654	2,503	2,393	2,287	2,207	2,095	-11%	-18%
Acetaldehyde	10,337	11,864	17,408	26,510	36,116	41,270	48,765	68%	249%
Formaldehyde	43,189	41,989	48,948	57,475	60,214	66,612	72,515	13%	39%
Benzene	6,902	6,633	6,285	6,185	6,134	5,866	5,677	-9%	-11%
PM2.5	5,686	5,049	4,412	3,831	3,253	2,640	2,000	-22%	-43%
PM10	29,526	26,217	22,908	19,893	16,893	13,708	10,388	-22%	-43%

Source: Faro90

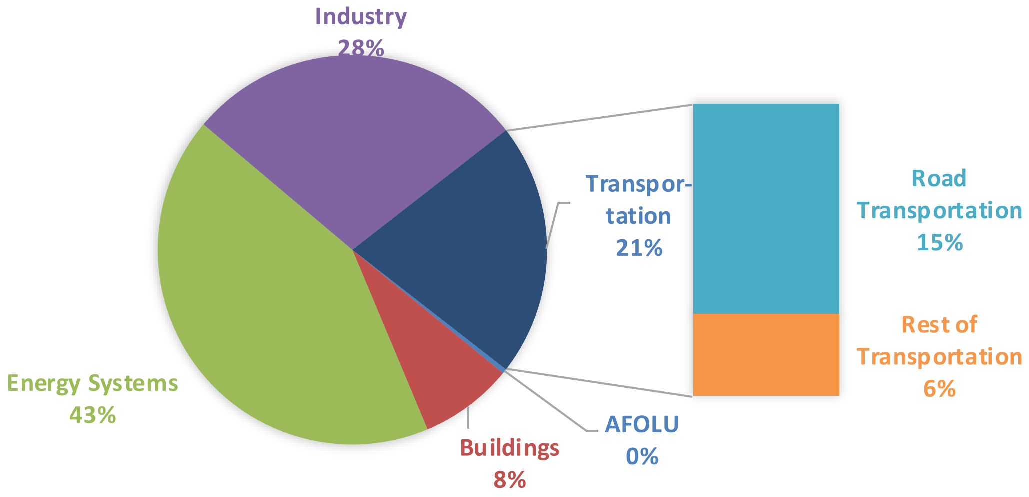
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

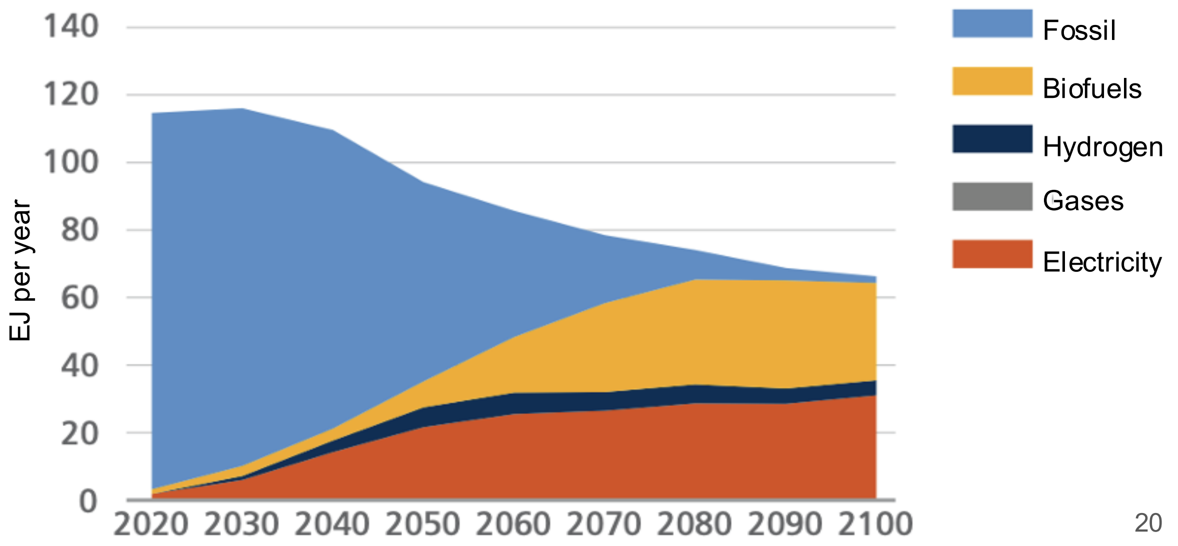
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

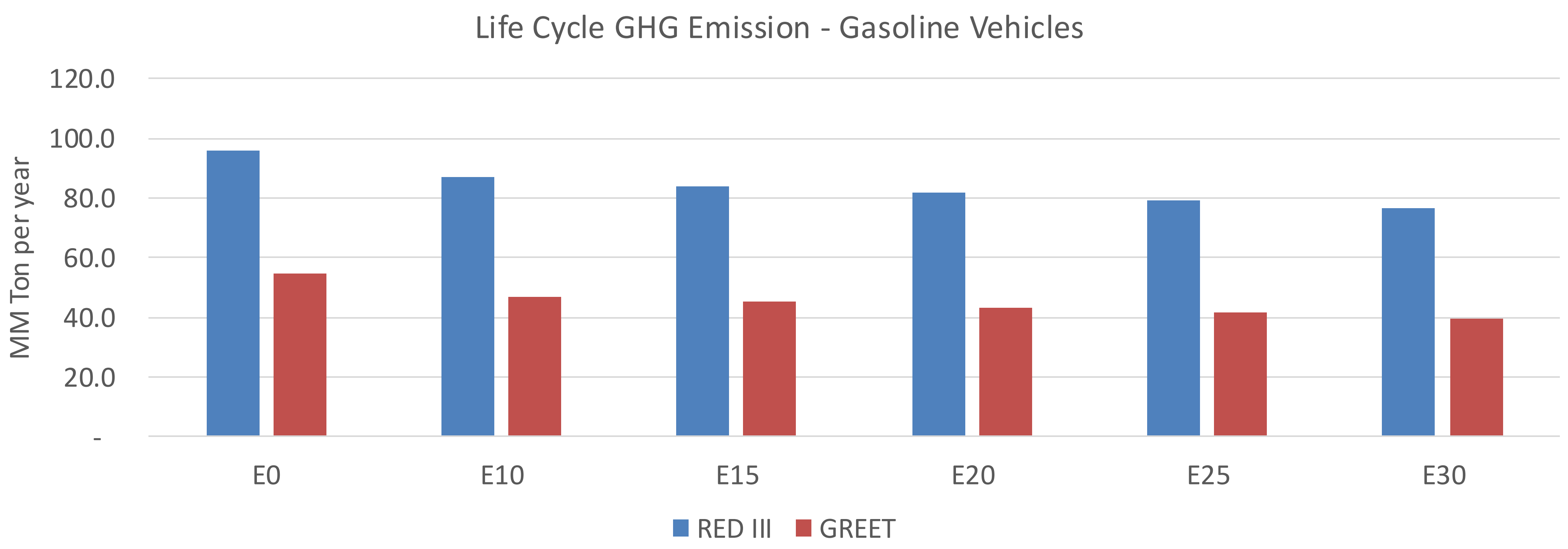
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Malaysia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	407.7
Target @2035 (MMT/year)	285.4
Delta	122.3

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	13.7%	17.4%	20.5%	23.4%	27.2%
RED II/III	0.0%	9.5%	12.3%	14.8%	17.2%	20.0%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	6.1%	7.8%	9.1%	10.5%	12.2%
RED II/III	0.0%	7.4%	9.6%	11.6%	13.5%	15.7%

Source: Greet, RED II/III, climateactiontracker.org, F90